Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CS5AI01/18DS5DE12]

FIFTH SEMESTER B.TECH EXAMINATION, DECEMBER–2021

ARTIFICIAL INTELLIGENCE

Time: 3 Hours **Maximum Marks:70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1. a. Sketch the Goal-based Agent structure and explain the various components of it. **07**

b. Describe the different environment types in which artificial agents can be deployed. **07**

2. a. Explain the steps in problem solving agent. **07**

b. Consider the figure 1 with five zones. Find an assignment of colours to zones such that no two adjacent zones are colored with the same colour. Apply the constraint satisfaction problem to colour the zones with minimum number of colours. Assume the colours [red, blue, green, yellow, pink] **07**

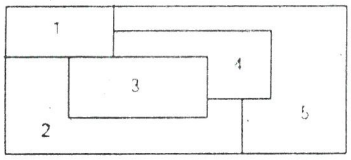

Figure 1

3. a. Consider the knowledge base models M1 and M2. **07**

M1: A, A→B and B M2: ~A, ~B and AvB. Conclude the following.

- **i.** [(A→B)∧A]→B is a tautology holds on M1
- **ii.** (AvB)∧[(¬A)∧(¬B)] is a contradiction holds on M2.

b. Describe the WUMPUS world problem and Identify the PEAS in the environment. **07**

OR

4. a. Consider the following statements about Your roommate comes home; he/she is completely wet. **07**

- **i.** Your roommate is wet
- **ii.** If your roommate is wet, it is because of rain, sprinklers, or both
- **iii.** If your roommate is wet because of sprinklers, the sprinklers must be on
- **iv.** If your roommate is wet because of rain, your roommate must not be carrying the umbrella
- **v.** The umbrella is not in the umbrella holder
- **vi.** If the umbrella is not in the umbrella holder, either you must be carrying the umbrella, or your roommate must be carrying the umbrella
- **vii.** You are not carrying the umbrella

Can you conclude that the sprinklers are on using rules of inferences?

b. Describe the syntax and semantics of predicate logic. **07**

5. a. Explain the steps involved in the Knowledge Engineering process for Planning. **07**

b. A spacecraft has to be launched for particular mission critical application. Can STRIPS planner be applied in this domain to plan the launch of the spacecraft? If not, provide suitable reasons. **07**

OR

6. a. Consider the blocks world problem as shown in the figure 2. Apply the STRIPS planner to solve the problem. **07**

Initial state Goal state

Figure 2.

b. Demonstrate the uses of hierarchical decomposition. **07**

7. a. Explain the components of performance elements to improve the design of learning agent. **07**

b. Discuss briefly Decision tree based learning. **07**

OR

8. a. Explain inductive learning algorithm. **07**

b. With a neat diagram, explain the single layer perceptron network. **07**